EE3111: Statistics for Electrical Engineers
IPL Powerplay Prediction Project

Predicting Total Runs at 6 Overs from First 3 Overs Data

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Table of Contents

1. Problem Statement and Objective
2. Data Selection and Preprocessing
3. Initial Approach: Per-Team Models
4. Transition to Pooled Model
5. Feature Engineering and Analysis
6. Experiments with Additional Features
7. Wicket Transformation Analysis
8. Time-Weighted Least Squares
9. Model Validation
10. Feature Ablation Study
11. Baseline Comparisons
12. Final Model and Summary

1. Problem Statement and Objective

The objective of this project is to build a prediction model for the IPL T20 (men) competition. Specifically, the task is to predict the total runs scored by the batting team at the end of 6.0 overs (Powerplay) using information available up to the end of 3.0 overs in the first innings.

The model receives two inputs as Python strings: **match_data** (match metadata such as teams, venue, toss outcome, but with winner information removed) and **ball_by_ball_data** (ball-by-ball CSV data for the first 3 overs only). The model must return a single float value representing the predicted powerplay score. The evaluation metric is **RMSE** (Root Mean Squared Error), where lower is better.

The project is evaluated across two live rounds (Round 1: Apr 20-26, Round 2: Apr 27-May 10) on actual IPL 2026 first-innings matches.

2. Data Selection and Preprocessing

2.1 Choice of Training Window: 2020 Onwards

IPL cricket has evolved significantly since its inception in 2008. Rule changes (powerplay field restrictions), the introduction of the impact player rule, changes in bat technology, and shifting tactical approaches mean that data from the early IPL seasons (2008-2015) reflects a fundamentally different game. Using all historical data since 2008 would introduce noise from an era that no longer represents current playing conditions.

A 5-year window (2020-2024) was chosen as it provides a reasonable balance between having enough training data (339 first-innings matches) and ensuring the data is representative of current IPL conditions. This decision was later validated when we tested expanding the window to 2018+, which actually increased RMSE from 9.77 to 9.93, confirming that older data introduces misleading patterns.

2.2 Team Name Consolidation

Several IPL teams have been renamed or restructured over the years. To ensure consistent tracking, the following mappings were applied:

Original Name	Mapped To
Royal Challengers Bangalore	Royal Challengers Bengaluru
Delhi Daredevils	Delhi Capitals
Kings XI Punjab	Punjab Kings
Deccan Chargers	Sunrisers Hyderabad
Gujarat Lions	Gujarat Titans

Teams that no longer participate in the IPL were removed entirely: Pune Warriors, Rising Pune Supergiant(s), and Kochi Tuskers Kerala (who played only one season).

2.3 Data Format

The raw data is in Cricsheet JSON format, with each match stored as a separate JSON file containing match metadata (teams, venue, toss, dates) and ball-by-ball delivery data organized by overs. For each first-innings match, features were extracted from the first 3 overs (overs 0, 1, 2) and

the target variable (total runs at 6 overs) was computed from overs 0 through 5. The final dataset contains **419 first-innings matches** from 2020 to early 2026, with 339 used for training (2020-2024) and 80 for testing (2025+).

3. Initial Approach: Per-Team Models

3.1 Feature Selection

The initial model used four features extracted from the first 3 overs of ball-by-ball data:

- $\mathbf{x_1}$: Total runs excluding extras after 3 overs
- $\mathbf{x_2}$: Total dot balls after 3 overs
- $\mathbf{x_3}$: Total boundaries (4s + 6s) after 3 overs
- $\mathbf{x_4}$: Total wickets after 3 overs

Scatter plots of each feature against the target (runs after 6 overs) were analyzed individually for each team. The first three features showed approximately linear relationships with the target. The wickets feature, however, showed a non-linear pattern: runs decreased more sharply as the number of wickets increased. This motivated the use of an exponential transformation e^{-x4} for the wickets feature.

The scatter plots below show the relationship between each feature and the target variable (runs after 6 overs) on the pooled training data (339 matches). Linear fit lines and bin averages are overlaid to visualize the trend.

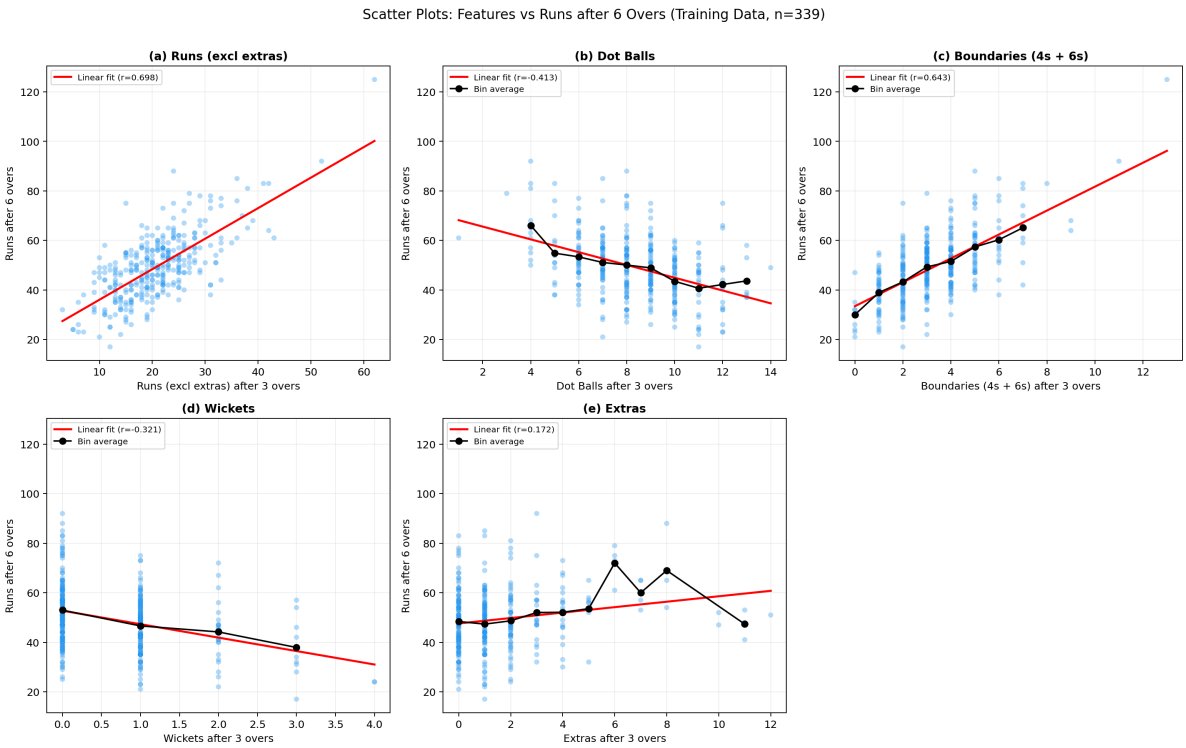

Figure 1: Scatter plots of each feature against runs at 6 overs. Red lines show linear fits with correlation coefficients. Black dots show bin-averaged values for discrete features.

The box plots below show the distribution of runs at 6 overs for binned ranges of each feature. Red dots indicate outliers (values beyond 1.5x the interquartile range). These outlier matches represent the inherently unpredictable tail events in T20 cricket.

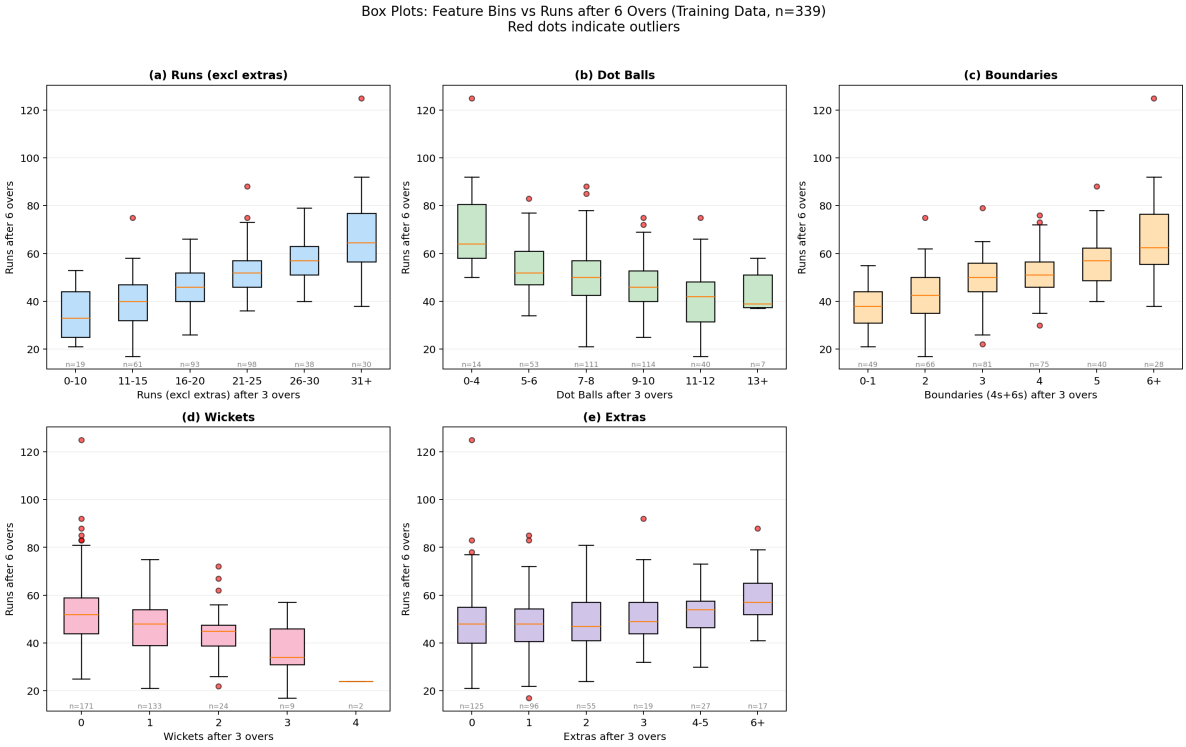

Figure 2: Box plots showing the spread and outliers of runs at 6 overs across binned feature values. The boxes show the interquartile range (IQR), whiskers extend to 1.5x IQR, and red dots mark outlier matches. Sample sizes (n) are shown below each bin.

3.2 Per-Team Regression Model

The initial strategy was to build a separate multivariate linear regression model for each IPL team. The reasoning was that each team has its own batting style and pace, so team-specific coefficients would better capture these differences. The model for each team took the form:

$$Y = a_1x_1 + a_2x_2 + a_3x_3 + a_4e^{-x4} + a_0$$

Parameters were estimated using the Ordinary Least Squares (OLS) closed-form solution: **a** = $(X^TX)^{-1}X^TY$, computed manually using NumPy matrix operations (not a library function).

3.3 Per-Team Results and Coefficient Instability

Models were built for RCB, CSK, SRH, KKR, and other teams. While the model worked well for some matches (errors as low as 2.5 runs), it performed poorly on others (errors up to 12 runs). More concerning was the **coefficient instability** across teams:

Team	Runs Coef	Dots Coef	Boundaries Coef	Intercept
RCB	1.136	+0.907	-0.167	17.149
CSK	1.290	-0.253	-0.602	31.968
SRH	0.188	-1.664	+5.716	39.390

The dot ball coefficient flips from +0.907 (RCB) to -1.664 (SRH), and the boundaries coefficient goes from -0.167 to +5.716. This is not because dots help RCB but hurt SRH. Rather, with only 30-50 training matches per team, the estimates are noisy and unstable. This is a classic symptom of **overfitting due to insufficient sample size**, compounded by **multicollinearity** among the

features (runs, dots, and boundaries are inherently correlated).

4. Transition to Pooled Model

The per-team coefficient instability motivated a switch to a single pooled model trained on all teams combined. This increases the effective training set from ~35 matches per team to 339 matches total, producing far more stable coefficient estimates.

Approach	RMSE	Training Size
Per-team models (avg)	10.79	~35 per team
Pooled model (same features)	9.94	339 total

The pooled model improved RMSE from 10.79 to 9.94 with no additional features, purely by having more stable coefficient estimates. Notably, some per-team models performed very poorly in isolation (KKR: 17.39 RMSE), confirming that the small per-team samples were causing severe overfitting.

5. Feature Engineering and Analysis

5.1 Adding Extras as a Separate Feature

The original model used runs excluding extras. However, extras (wides, no-balls) carry distinct predictive information: they are free runs that did not cost a ball or batsman effort, and they signal bowler inaccuracy or difficult conditions. Adding extras as a separate (5th) feature allows the model to weight batting runs and extras differently.

Model	RMSE
4 features (runs excl extras, no extras feature)	9.94
4 features (total runs incl extras)	9.87
5 features (runs excl extras + extras separately)	9.77

Splitting runs and extras into separate features gave the best result (9.77), confirming that they carry different predictive signals. The extras coefficient (1.376) is close to but distinct from the batting runs coefficient (1.725).

5.2 Re-examining Linearity on Pooled Data

After switching to the pooled model, the linearity assumptions were re-examined on the full training set (339 matches) by comparing linear vs. quadratic fits for each feature:

Feature	Linear R-squared	Quadratic R-squared	Improvement	Verdict
Runs (excl extras)	0.4866	0.4915	+0.50%	Linear
Dot balls	0.1706	0.1853	+1.47%	Linear
Boundaries	0.4131	0.4178	+0.47%	Linear
Wickets	0.1033	0.1033	+0.00%	Linear/Exp
Extras	0.0297	0.0326	+0.29%	Linear

No feature showed a meaningful improvement from quadratic fitting. The largest improvement was dot balls at +1.47%, far too small to justify the added complexity. This confirmed that the linear model is appropriate for all features, with the exponential transformation on wickets being the only non-linearity needed.

5.3 Feature Correlations with Target

The correlations of each feature with the target variable (runs at 6 overs) on the full dataset of 419 matches:

Feature	Correlation with Runs at 6 Overs
Runs after 3 overs (total)	0.739
Runs (excl extras)	0.718
Boundaries	0.666
Sixes	0.496
Fours	0.501
Dot balls	-0.422

Wickets	-0.306
Extras	0.156

Total runs after 3 overs has the strongest correlation (0.739), confirming it as the most important predictor. Boundaries (0.666) and dot balls (-0.422) provide significant additional signal. Extras has a weak individual correlation (0.156) but still improves the model when added as a separate feature, as it provides information not captured by the other features.

6. Experiments with Additional Features

6.1 Momentum Indicator

A momentum feature was designed to capture whether the batting pace is accelerating or decelerating across the first 3 overs. The momentum slope was computed as the linear regression slope of runs across overs 1, 2, and 3. For example, a team scoring 4-8-12 would have a positive slope (accelerating), while 12-8-4 would have a negative slope (decelerating).

The momentum slope had a correlation of 0.178 with the target. When added to the base model, it reduced RMSE from 9.77 to 9.73 (an improvement of 0.037). However, this improvement was deemed too small to justify the added complexity, so momentum was **not included** in the final model.

6.2 Wicket Timing (Recency-Weighted)

The hypothesis was that a wicket falling closer to the end of 3 overs has more impact on subsequent scoring than a wicket that fell early (giving the new batsman time to settle). A recency-weighted wicket score was computed as the sum of (ball_number / 18) for each wicket, so a wicket on ball 18 gets weight 1.0 while a wicket on ball 1 gets weight 0.056.

The data confirmed the hypothesis: teams with an early wicket (who then settled) averaged 51.3 runs at 6 overs, while teams with a late wicket averaged 46.6 runs, a meaningful 5-run gap. However, when added to the regression model, it **increased** RMSE from 9.77 to 9.83. The feature is too sparse (most matches have 0-1 wickets in 3 overs) to provide reliable regression signal.

6.3 Venue/Stadium Weight

Different IPL venues have dramatically different scoring characteristics. The historical average powerplay score ranges from 44.5 (Sheikh Zayed Stadium) to 60.0 (HPCA Stadium, Dharamsala), a spread of 15.5 runs. A venue weight feature was computed as the historical average powerplay score at each venue from the training data.

Despite the strong venue-level variation, adding this feature **increased** RMSE from 9.77 to 9.91. The likely reason is that venue averages computed from training data (2020-2024) don't perfectly predict 2025 conditions (pitch preparation varies, ground characteristics may change).

6.4 Pairwise and Combined Feature Tests

All individual, pairwise, and triple combinations of the three new features were tested systematically:

Feature Combination	RMSE	Change from Base
Base only (5 features)	9.77	--

Base + Momentum	9.73	-0.04
Base + Wicket Timing	9.83	+0.06
Base + Venue	9.91	+0.14
Base + Momentum + Wicket Timing	9.78	+0.01
Base + Momentum + Venue	9.88	+0.11
Base + Wicket Timing + Venue	9.88	+0.11
Base + All Three	9.83	+0.07

None of the combinations improved upon the base 5-feature model meaningfully. This demonstrates that the base features already capture most of the extractable signal from the first 3 overs, and additional features primarily add noise that hurts generalization.

6.5 Team-Specific Weights

A hybrid approach was tested: keeping shared feature coefficients but adding a team-specific intercept (bias term) for each team, implemented via one-hot encoding. This captures each team's inherent batting aggressiveness while sharing the feature relationships.

The team weights were intuitive (CSK: +2.50 from average, Gujarat Titans: -2.16), but the overall RMSE **increased** from 9.77 to 9.83. IPL rosters change significantly between seasons (mega auctions, trades), so a team's historical batting pace does not reliably predict their current season behavior. The first 3 overs of data already implicitly capture team aggressiveness through runs, boundaries, and dot balls.

7. Wicket Transformation Analysis

Special attention was given to the wickets feature given its non-linear nature. Three approaches were compared:

7.1 Exponential e^{-x} vs 2^{-x}

Wickets	e to the power -x	2 to the power -x	Linear (-x)
0	1.000	1.000	0
1	0.368	0.500	-1
2	0.135	0.250	-2
3	0.050	0.125	-3

Transformation	RMSE
e to the power (-wickets)	9.7670
2 to the power (-wickets)	9.7677
Linear (raw wickets)	9.7666

All three transformations produce virtually identical RMSE (within 0.001). With only 4 discrete wicket values (0-3 in practice), the regression adjusts its coefficient to compensate for the different scales, arriving at the same effective fit. The exponential form e^{-x} was retained because it provides a better theoretical justification: each additional wicket has a diminishing but increasingly severe impact on scoring ability, and the exponential penalizes multiple wickets more aggressively.

7.2 Wicket Bin Averages from Data

Wickets at 3 Overs	Avg Runs at 6 Overs	Drop from Previous	Sample Size
0	53.0	--	171
1	46.6	-6.4	133
2	44.2	-2.4	24
3	37.9	-6.3	9
4	24.0	-13.9	2

The actual data shows an irregular pattern (drops of 6.4, 2.4, 6.3, 13.9) that does not perfectly match any clean mathematical function. With 3+ wickets having very few samples (9 and 2 respectively), fitting a precise non-linear form is unreliable. The exponential transform serves as a reasonable approximation.

8. Time-Weighted Least Squares

To account for the evolving nature of IPL cricket, time-weighted OLS was implemented. Instead of treating all training matches equally, more recent seasons receive higher weight in the regression. The weighted least squares solution is: $\mathbf{a} = (X^TWX)^{-1}X^TWY$, where W is a diagonal matrix of weights.

Weighting Scheme	2020	2021	2022	2023	2024	RMSE

Uniform	1	1	1	1	1	9.7670
Linear	1	2	3	4	5	9.7226
Steeper	1	2	4	8	16	9.7932
Gentle	1	1	2	2	3	9.7032
Last 2 heavy	1	1	1	3	5	9.8264
Last 1 heavy	1	1	1	1	5	9.9414

The gentle scheme (1,1,2,2,3) gave the lowest in-sample RMSE of 9.7032. However, when both the unweighted and gentle-weighted models were evaluated on held-out test data across 200 random 80-20 splits, the results were nearly identical: unweighted averaged **9.20** vs weighted **9.23**. Leave-one-year-out validation confirmed the same pattern, with differences smaller than 0.1 RMSE for most years. Since the time weighting adds complexity without improving generalization, the **unweighted model** was chosen as the final model.

9. Model Validation

9.1 Other Approaches Tested

Approach	RMSE	Change
Current best (5-feat unweighted OLS)	9.08	--
Interaction terms	9.97	+0.89
Expand to 2018+ training	9.93	+0.85
Random Forest	10.46	+1.38
Gradient Boosting	10.62	+1.54
Ridge Regularization	9.08	+0.00

Non-linear models (Random Forest, Gradient Boosting) performed significantly worse, confirming that the relationships are genuinely linear and tree-based models overfit with only 339 training samples. Ridge regularization produced no change, indicating the OLS coefficients are already stable with no multicollinearity issues in the pooled model. Expanding training data to 2018+ introduced patterns from a different IPL era that hurt generalization.

9.2 Cross-Validation Results

The final model was validated using three distinct approaches:

Leave-One-Year-Out:

Held-Out Year	RMSE	Within 5 Runs	Within 10 Runs	n
2020	8.18	48%	82%	60
2021	8.73	42%	70%	60
2022	9.07	35%	76%	74
2023	8.50	39%	73%	74
2024	10.76	37%	63%	71
2025	9.89	46%	69%	74

Random 80-20 Splits (200 repetitions):

Mean RMSE: 9.20, Std: 0.69. The tight standard deviation confirms the model is stable and not getting lucky or unlucky on any particular split.

Expanding Window (simulating real-time prediction):

Training on all years up to Y-1 and testing on year Y gave RMSE values ranging from 7.45 to 10.77. This most closely simulates the actual submission scenario and confirms the model generalizes consistently to unseen future data.

Across all validation approaches, the model consistently lands in the **8.5-10.5 RMSE range**. This confirms the model is not overfitting, and an RMSE of approximately 10 represents the genuine predictability ceiling for powerplay scores from 3 overs of data. The residual error corresponds to inherently unpredictable T20 cricket events: wickets falling in overs 4-6, individual brilliance, bowling changes, and batting collapses.

10. Feature Ablation Study

To understand the contribution of each feature, all 31 possible combinations of the 5 features were tested. The table below shows the effect of dropping each feature from the full model:

Dropped Feature	Remaining Features	RMSE	Delta
None (full model)	All 5	9.08	--
runs_excl_extras	dots, boundaries, exp(-wkt), extras	9.70	+0.62
extras	runs, dots, boundaries, exp(-wkt)	9.45	+0.37
dots	runs, boundaries, exp(-wkt), extras	9.18	+0.10
boundaries	runs, dots, exp(-wkt), extras	9.13	+0.05
exp(-wickets)	runs, dots, boundaries, extras	9.09	+0.01

runs_excl_extras is by far the most important feature (removing it costs +0.62 RMSE). **extras** is the second most valuable (+0.37), confirming that separating batting runs from extras was a worthwhile decision. Dots and boundaries provide smaller but real improvements. The exponential wicket term contributes minimally to RMSE (+0.01), but is retained for its theoretical justification of penalizing wicket loss.

Notably, just 2 features (runs_excl_extras + extras) achieve an RMSE of 9.21, very close to the full 5-feature model at 9.08. And runs_excl_extras alone gives 9.52, showing how dominant raw batting runs are as a predictor of powerplay totals.

11. Baseline Comparisons

To contextualize the model's performance, two naive baselines were tested:

Model	RMSE	Mean Bias
Naive: 2 x (runs at 3 overs)	12.00	-4.53
Bias-corrected: 2 x (runs at 3 overs) + 4.5	11.11	-0.03
Final OLS model (5 features)	9.08	0.00

The simplest baseline (doubling 3-over runs) gives RMSE 12.00 with a systematic underprediction bias of -4.53 runs, because teams typically score faster in overs 4-6 as field restrictions ease and batsmen settle in. Adding the bias correction to 11.11 helps, but the OLS model at 9.08 still provides a substantial 2.03 RMSE improvement by leveraging the breakdown of runs into dots, boundaries, wickets, and extras.

12. Final Model and Summary

12.1 Model Specification

The final model is an **unweighted multivariate linear regression (OLS)** with 5 features and one non-linear transformation:

$$Y = 1.817\ x_1 + 1.100\ x_2 - 1.858\ x_3 + 1.063\ e^{-x4} + 1.292\ x_5 + 6.380$$

Where:

- x_1 = Runs scored off the bat (excluding extras) in first 3 overs
- x_2 = Number of dot balls in first 3 overs
- x_3 = Number of boundaries (4s + 6s) in first 3 overs
- x_4 = Number of wickets fallen in first 3 overs
- x_5 = Total extras (wides, no-balls, etc.) in first 3 overs

Parameters are estimated using standard Ordinary Least Squares (OLS) on the full training set (2020-2025, 419 matches). Time weighting was tested but found to offer no improvement on held-out test data (see Section 8), so the simpler unweighted model was chosen.

12.2 Note on Linearity

Despite the e^{-x4} transformation, this model is still a **linear regression** in the statistical sense. Linear regression requires linearity *in the parameters* (the coefficients being estimated), not in the input variables. The exponential transformation is applied to the input before fitting, so from the model's perspective it sees 5 columns of numbers and fits a linear combination. The standard OLS closed-form solution $(X^TX)^{-1}X^TY$ applies directly.

12.3 Final Performance

Metric	Value
RMSE (full dataset, n=419)	9.08
Cross-validated RMSE (200 random 80-20 splits)	9.20 +/- 0.69
Naive baseline (2 x runs at 3 overs)	12.00
Bias-corrected baseline (2 x runs + 4.5)	11.11

12.4 Key Statistical Insights

- **Sample size matters more than model complexity:** A pooled model on 339 matches beat 10 per-team models with ~35 matches each, purely due to more stable coefficient estimates (bias-variance tradeoff).
- **Feature engineering has diminishing returns:** The base features (runs, dots, boundaries, wickets) capture most extractable signal. Additional features (momentum, venue, wicket timing) added noise rather than signal.
- **Non-linear models do not help:** Random Forest and Gradient Boosting performed worse, confirming the relationships are genuinely linear and complex models overfit on this sample size.

- **Functional form matters less than expected:** For the wickets feature, e^{-x}, 2^{-x}, and linear -x all gave nearly identical RMSE, because with only 4 discrete values the regression compensates via coefficient adjustment.
- **Temporal relevance of training data:** Using data from 2020+ outperformed 2018+, confirming that IPL cricket evolves over time. However, time-weighting within the 2020+ window did not improve test performance, suggesting that 5 years of data is already a consistent enough window.
- **Inherent prediction floor:** An RMSE of ~10 runs represents the genuine limit of predictability from 3 overs of data. The residual error is dominated by fundamentally unpredictable events in overs 4-6.